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Proposal Title:	A system approach to alcohol policy evaluation: using microsimulation to model the effect of policy changes on alcohol use and related harm

ABSTRACT

Alcohol use is one of the main contributors to the global disease burden. Alcohol use is causally related to over 200 chronic and acute health conditions, including several types of cancer, cardiovascular, gastrointestinal, injury, and neuropsychiatric disorders. Alcohol use disorder by itself is among the most prevalent psychiatric disorders, with a global estimated prevalence of 5.1% (95%CI:4.9, 5.4%). All of this has a correlation with other dimensions of society, including economic wellbeing, public and domestic safety, and social development. In Chile, the latest estimates indicate that approximately 13% of all deaths that occurred in 2014 were causally related to the use of alcohol; its economic costs were estimated to be 0.8% of the gross domestic product, equivalent to US\$2,24 billions in 2017. The enormous health and social burden of alcohol consumption warrants urgent and comprehensive approaches to address this problem.

Despite the above information, the design and implementation of policy solutions are still lacking, in part because of the difficulties in evaluating the population impact of regulatory changes occurred in the past that justify further reforms, but also because of strong lobbies from the alcohol industry in countries such as Chile. One possible response to this problem is the use of **current computational advances to simulate real-world scenarios** (e.g., dynamics of alcohol use and related harm) affected by different exogenous inputs (e.g., policy changes) using the best available information and evidence from decades of scientific research. Results from studies that use a complex system perspective can produce reliable information on population dynamics in alcohol use, as well as the potential impact of policy changes under different scenarios and assumptions, which can inform decision-making for the design and implementation of effective interventions.

In this project, **we aimed to (1)** Select and describe specific alcohol policies based on their effectiveness, expected results, and feasibility of implementation in Chile in the next decade; **(2)** generate a simulation model that mimics the empirical distribution, trend, and dynamics of alcohol use and related harm in the population of 15+ years of age of Chile between 2008 and 2019; and **(3)** estimate the 10-year impact of selected alcohol policies on alcohol use, cause-specific alcohol-related mortality, and years of potential life lost, across different demographic and socioeconomic groups, through a simulation model of the Chilean population.

Because it is impractical to simulate the effect of all alcohol policies in this project, for **aim 1**, we will use the World Health Organization's "best buys" to initiate a guided discussion with experts and key informants to **select and design three specific policy changes** that will be used in the simulation model. To do this, we will use the Delphi method to rank policies using the following inputs for discussion: 1) cost-effectiveness in a social and epidemiological context like Chile, and 2) feasibility of being implemented in a 10-year period in the Chilean context. Then, we will ask informants about the key features and assumptions of the selected policies to incorporate that knowledge in model building. For **Aims 2 and 3, we will generate a microsimulation model** using individual-level data to capture socio-demographic characteristics, behavioral patterns, and contextual information on alcohol use and related harms. We will use data and information from different sources, including census data, national surveys, administrative information, and mortality to specify behavioral patterns and decision-making algorithms for individuals in different subgroups of a synthetic population under different policy scenarios and assumptions (informed by aim 1). The microsimulation will be initialized and structured around a set of mutually exclusive discrete states (e.g., category of alcohol use), from which individuals will have a known probability of transitioning (aim 2). The current scenario (dynamics of alcohol use and related harms under the present regulatory scheme) will be projected for the next 10 years and contrasted with different counterfactual scenarios to quantify the simulated impact of policy changes (aim 3).

Our proposal advances several areas of scientific research, including using for the first time a microsimulation model to estimate the effects of alcohol policies in Chile and Latin America through a **participatory, interdisciplinary** (computer science, economics, public policy, public health, and epidemiology), **and a system approach**. Our team has a strong publication record and years of experience in conducting research on alcohol use and related harm, generating primary data, and using secondary information to evaluate and simulate policy effects. The protocol of our will be pre-registered, and all the analytical tools and scripts generated will be posted in open repositories to allow replication and adaptation to respond to other research questions.

A. THEORETICAL FOUNDATIONS AND STATE OF THE ART

Alcohol use is among the leading risk factors for morbidity and mortality worldwide. In Chile, approximately 13% of all deaths that occurred in 2014 were causally related to the use of alcohol;¹ among young men, alcohol accounts for up to 50% of mortality.² The enormous health and social burden of alcohol consumption warrants urgent and comprehensive approaches to address this problem. Complex system models can be a valuable and cost-effective alternative to understand the dynamics of social and health problems and estimate the effects of hypothetical interventions across different subgroups of the population. Following prior advances in this field,³⁻⁵ we propose to generate a simulation model that mimics the empirical distribution, trend, and dynamics of alcohol use and related harm in the Chilean population, so that it can be used to estimate the impact of hypothetical policies on alcohol use, cause-specific alcohol-related mortality, and years of potential life lost, across sociodemographic groups and through a projected time span of 10 years. To achieve this, we will generate a microsimulation model using individual-level data on the distribution of sociodemographic characteristics in the Chilean population, behavioral patterns, and contextual information on alcohol use and related harm. Data and information will be collected from different sources, including social and alcohol surveys, longitudinal data on social dynamics, and mortality data. The microsimulation will be structured and calibrated using real-world scenarios (e.g., dynamics of alcohol use and related harm) and different exogenous inputs (e.g., policy changes). Outputs include effect measures in the absolute and relative scales for the entire population and across specific subgroups. Our project will advance several areas of scientific research, including using a microsimulation model for the first time to estimate the effects of alcohol policies in Chile and Latin America through a participatory and interdisciplinary approach. Our results are expected to inform public policy discussion by providing reliable estimates of plausible policy changes. Our protocol will be pre-registered and model codes will be available to allow reproducibility and extension to other policies and outcomes.

A1. The epidemiology of alcohol use

Alcohol use is one of the main contributors to the burden of disease globally.⁶ It is causally related to over 200 chronic and acute health conditions, including several types of cancer, cardiovascular, gastrointestinal, and infectious diseases.⁷ Alcohol use is also a main contributor to the burden of neuropsychiatric conditions. Alcohol use disorder (AUD) by itself is among the most prevalent psychiatric disorders, with a global estimated prevalence of 8.6% (95%CI: 8.1, 9.1%) for men and 1.7% (95%CI: 1.6, 1.9%) for women in 2016, with a total estimated prevalence of 5.1%; (95%CI: 4.9, 5.4%).⁸ Likewise, alcohol use disorder is highly disabling and is associated with multiple physical and psychiatric comorbidities;⁹ it is estimated that it accounts for approximately 10% of the burden of disease related to substance use and mental disorders.¹⁰ Alcohol consumption, particularly episodes of heavy drinking, is also related to several external causes of morbidity and mortality, including motor vehicle accidents, falls, drowning, poisonings, as well as intentional self-injuries and harm to others.¹¹⁻¹⁴ Indeed, the harm to other from drinking has gained attention in the past decade, not because of its novelty but because of the need to quantify the true magnitude of the problem across different societies, contexts, and population subgroups.¹²

All the above has a correlation with other dimensions of society. Economic well-being, public and domestic safety, and social development are also dimensions affected by alcohol use. Studies on the economic costs of alcohol use systematically show that the net cost of alcohol use is significant, ranging from 1.2% to 1.7% of the total gross domestic product.¹⁵ These include direct cost in healthcare, AUD treatment, medical leave, indirect costs from loss of productivity due to work absenteeism and premature death, and costs from the judicial system.^{1,16} Multiple forms of violence are strongly associated with alcohol use, most notably domestic violence. Studies have shown that between 25% and 70% of reported episodes of domestic violence are related to the use of alcohol,¹⁷ with women and children being the overwhelming majority affected in these cases.¹⁷ But the impact of drinking does not have an equal distribution in the population. As for many other social and health problems, **it mostly affects low- and middle-income groups**, despite having a larger prevalence of use among people from most affluent areas.^{18,19} The latter is known as the alcohol harm paradox and can be explained by two main factors: 1) the larger prevalence of heavy drinking episodes and alcohol use disorders among drinkers in lower socioeconomic groups and 2) fewer resources to cope with problematic alcohol use and associated harms, including access to treatment and healthcare services.¹⁸ This results in the amplification of existing social inequities, which reaffirms the need and urgency of tackling this problem from a comprehensive public health perspective.

A1.1 The epidemiology of alcohol use in Chile

Chile is the country in the Americas with some of the larger volumes of per capita alcohol use.²⁰ According to the last National Household Surveys on Drug Use of Chile including people between 12 and 64 years of age, the past month prevalence of alcohol use was 44.3%, with little variation across socioeconomic groups

and sex.²¹ That same study showed that 1 out of 2 people who drank in the past month reported at least one episode of binge drinking in that same period (i.e., 5+ standard drinks in a single occasion for men and 4+ for women), with significant differences across sexes (55.2% in men vs. 43.3% in women) and socioeconomic group (54.4% in low socioeconomic groups vs. 45.5% in high socioeconomic groups).²¹

There are no exact estimates on the number of people in treatment for alcohol use disorder in Chile, though the National Service for the Prevention and Rehabilitation of Drugs and Alcohol Consumption (SENDA) funds approximately 16,500 treatments per year for people 18+ years of age and about 35% (n≈10,700 are for people declaring alcohol as their main problematic substances. To that number, it can be added the number of treatment episodes with people declaring alcohol as a secondary or contributory substance (n≈7,500), as well as people in treatments funded by the Ministry of Health (n≈10,000 for all drugs) and those in private out-of-pocket treatment (data unknown). However, it is assumed that treatment needs in the population are substantially higher than the treatment offer, with a potential demand for all drugs of approximately 110,000 people, of which 69% are alcohol-related, according to SENDA's estimates.²² These estimates are for the general population of 18+ years of age and do not consider special programs for people in prison, parole, or homelessness. There are other public efforts to cover other needs in the population, including targeted prevention program or screening, brief intervention, and referral to treatment programs, most of them installed in primary care or school settings.²² For example, it is estimated that each year approximately 35,000 people are screened and receive a standardized brief interventions or basic treatment in primary care; the impact of this programs is, however, still unknown.²³

To our knowledge, three studies have estimated alcohol-attributable mortality in Chile.^{1,2,24} All of these studies have similar methodologies, which are based on the estimation of the alcohol population attributable fraction to calculate the number of deaths that are totally or partially attributable to alcohol use. The first study was the Chilean Burden of Disease Study, in which the number of deaths attributable to alcohol use was 8,366 in 2004, corresponding to 9.7% of the total deaths for that year.²⁴ Consistently, in a subsequent study using mortality data for the year 2009 and a more refined calculation of the population attributable fraction, the estimated number of deaths was 8,750, corresponding to 9.8% (95%CI:7.0%, 13.0%) of all deaths in that year.² In the most recent study on the subject, the authors estimated that 13% of the deaths in 2014 were attributable to alcohol use, corresponding to 13,260 dead cases that year.¹ But even as impressive as these numbers are, they hide the real impact of alcohol use in some population subgroups. For example, among young (15-29 years of age) men, alcohol use is responsible for up to 50% of all deaths, explained mostly by the large proportion of deaths from external causes, particularly motor vehicle accidents, homicides, and suicides.² In this sense, years of potential life lost is a sensible measure that includes the extent to which alcohol impacts younger age groups.

Other negative externalities associated with the use of alcohol are harm to others from drinking and its economic costs. Most of what we know about harm to others in Chile comes from a survey conducted in 2013 among 18+ years of age, following the WHO/Thai Health protocol.^{25,26} Results of that study showed that 1 out of 3 respondents reported that the effect of alcohol consumption by third parties was somewhat or very negative, being the most frequent negative effects verbal abuse (37.4%), being bullied (17.4%), feeling fearful at home or in a private meeting (19.6%), and family problems (16.9%).²⁵ Regarding the economic costs, the most updated study in Chile was conducted in 2018 using data from 2017 to calculate the economic costs.¹ The study considered economic costs related to alcohol from health care, violence and crime, productivity loss, and several social programs implemented at different levels (national, municipal) by different institutions.¹ The study showed that 0.8% of the gross domestic product, equivalent to US\$2,24 billions in 2017 were related to the use of alcohol; of these, 30.1% were due to health related costs and 52.2% due to loss of productivity. The authors, however, warn about the serious underestimation of productivity loss due to the lack of good quality data, so the total cost is expected to be significantly higher.

A2. Policy approaches to prevent alcohol use and related harm

There is a large set of policies designed to prevent alcohol consumption and its negative externalities. The World Health Organization (WHO) grouped policies into 10 domains: 1) pricing policies; 2) regulating marketing; 3) availability of alcohol; 4) drink-driving policies and countermeasures; 5) health services' response; 6) reducing the negative consequences of drinking and alcohol intoxication; 7) reducing illicit alcohol and informally produced alcohol; 8) community action; 9) leadership, awareness, and commitment; and 10) monitoring and surveillance. Within these domains, some policies are described as "best buy" because of their known cost-effectiveness in reducing alcohol and related harm (See **Section D4.2.** for additional details).²⁰ Each of these 10 domains contains between 3 and 9 specific policies, and each one of those can have multiple possible designs. For example, alcohol pricing domain groups initiatives related to increasing/revising alcohol taxation, banning or restricting price promotions or discount sales, establishing

minimum prices for alcohol, and reducing or stopping subsidies to economic operators in the area of alcohol, among others.

In Chile, there have been some changes in the direction suggested by the WHO, although legislators are still reluctant to adopt stricter regulations.²³ Law No. 19,925 of 2004,²⁷ for example, regulates the sale of alcoholic beverages, their labeling, and regulations on advertising. It also considers measures to prevent and rehabilitate problematic consumption. This law was recently modified in 2021 by Law No. 21,363,²⁸ particularly regarding marketing and advertising of alcoholic beverages in Chile. Among the most relevant modifications introduced was the prohibition of the sale of alcoholic beverages in gasoline stations, the obligation to include clear warnings on labels about the risks and harm associated with alcohol consumption (still not implemented due to controversies in the bylaw), and the prohibition of advertising in sports spaces or at times between 6 a.m. and 10 p.m. In addition, to restrict access to alcohol by minors, this law requires showing an identity card or other valid identification documents to all persons who wish to purchase alcoholic beverages. To the best of our knowledge, no studies have evaluated the implementation, enforcement, or effects of these legal changes.

Another example in Chile is Law No. 20,780 enacted in 2014 ("Tax reform that modifies the income tax system and introduces various adjustments to the tax system"),²⁹ which introduced modifications to Decree Law No. 825 of 1974, including modifications to income tax from sale and services, including those corresponding to alcoholic beverages. In Article 2, letter b), it is established that "Liquors, piscos, whiskey, spirits, and distillates, including liqueur or flavored wines similar to vermouth, [will have an] additional [tax] rate of 31.5%.". Letter c) of the same article states: "Wines intended for consumption, including carbonated wines, sparkling wines or champagne, generous or sunny wines, chichas, and ciders intended for consumption, regardless of their packaging, beer, and other alcoholic beverages, whatever their type, quality or denomination, [will have an] additional [tax] rate of 20.5%". This corresponds to a 3.5 percentage points increase to the additional tax rate in place prior to 2014 (changed from 27% to 30.5%) for alcoholic beverages included in letter b) and 5.5 percentage points (from 15% to 20.5%) for beverages included in letter c). It should be noted that the tax rates mentioned in the law are additional to the value-added tax, which is 19% in Chile, and additional to customs duties of 6% (*ad valorem*). In other words, the national treasury collects the equivalent of 39.5% for beer and 50.5% for spirits as a tax on alcoholic beverages produced in Chile. In comparative terms, excise taxes in Chile are above the median in Latin America, which is 21.4% for spirits, 11% for wine, and 13.8% for large beer (750ml).³⁰ However, from a public health perspective our tax scheme (*ad valorem*) incentivizes consumers to prefer cheaper brands and potentially alcohol from illegal or informal sources.

A3. Research gaps

Despite the importance of the problem, **the design, implementation, and evaluation of alcohol policy changes are still lacking**, particularly in countries in the Global South. This is likely because of the difficulties in evaluating the population impact of regulatory changes that occurred in the past that justify further reforms, but also because of strong lobbies from the alcohol industry in countries such as Chile. In 2021, Peña et al. published a compelling review of alcohol policy proposal and impact evaluation of policies in Chile and concluded that policies with the highest likely cost-effectiveness had not been implemented and only the impact of drink-driving policies has been evaluated.²³

One possible solution to the lack of evidence on alcohol policy is the use of current computational advances to simulate real-world scenarios (e.g., dynamics of alcohol use and related harm) under different exogenous inputs (e.g., policy changes) using the best available information and evidence from decades of scientific research. **Microsimulations are increasingly used in modeling the complex dynamics of public health phenomena** as well as in the investigation of causal mechanisms and intervention effects.³ This approach allows for the investigation of behaviors and outcomes at an individual level, producing outputs across different groups in the simulated (synthetic) population, incorporating dynamic demographic (e.g., ageing), social (e.g., socioeconomic mobility), and health processes (e.g., mortality).³

As shown in prior publications, results from studies that use a **complex system perspective can produce reliable information on population dynamics in alcohol use**, as well as the potential impact of policy changes under different scenarios and assumptions, which can inform decision-making for the design and implementation of effective interventions.^{3-5,31-33} Example of large simulation models include the Sheffield Alcohol Policy Model,⁵ the Calibrated Agent Simulations for Combined Analysis of Drinking Etiologies (CASCADE) model,³ and the Simulation of Alcohol Control Policies for Health Equity (SIMAH) project.⁴ The Sheffield Alcohol Policy Model was designed to provide estimates of the effectiveness and cost-effectiveness of alcohol policies in the UK. It is probably one of the most influential simulation models, with several adaptations and extensions to include additional alcohol policies to the one originally considered (pricing and availability policies, as well as screening and brief interventions). The primary aim of the CASCADE model

was to develop a simulation model of alcohol use based on existing theories explaining individual drinking and seeking novel combinations of these theories to better explain the changes in alcohol use observed in society. For example, this model uses a component of the Theory of Planned Behavior and Dual Process Theories to model individual decision processes in alcohol.³⁴ The SIMAH project aim to investigate the extent to which alcohol use affects mortality rates underlying the recent stagnation and decline in US life expectancy and the increasing inequalities in life in the US.

According to a 2021 scoping review of studies applying a complex systems perspective to alcohol consumption, there are no studies using this approach in Latin America, and only 3 out of 87 publications were from non-high-income countries (South Africa and Nepal); none of those three include a policy evaluation component.³⁵

B. RESEARCH QUESTION AND HYPOTHESIS

Our main **research questions** are as follows: According to experts and key informants, what alcohol policies are the most effective and feasible to implement in Chile in the next decade? What is the simulated 10-year impact of the selected alcohol policies on alcohol use, cause-specific alcohol-related mortality, and years of potential life lost in the Chilean population?

We **hypothesize that** 1) the existing data on social dynamics, alcohol use, and mortality in Chile are sufficient to build a microsimulation model that mimics the empirical distribution, trend, and dynamics of alcohol use and related harm in the population of Chile between 2008 and 2019, and 2) this microsimulation model can be used to estimate the 10-year impact of selected alcohol policies on alcohol use, cause-specific alcohol-related mortality, and years of potential life lost in the Chilean population of 15+ years of age. Alcohol policies and their specific provisions will be established inductively from expert elicitation, and they are expected to have a protective effect on alcohol use and related harm in quantities consistent with the scientific literature.

C. GOAL

The goal of the present project is to generate a simulation model that reproduces the empirical distribution, trend, and dynamics of alcohol use and related harm in the population aged 15+ years of Chile, which can be used to estimate the impact across different demographic and socioeconomic groups of hypothetical alcohol policies on a range of behavioral and health outcomes, namely alcohol use, cause-specific alcohol-related mortality, and years of potential life lost, for the next 10 years. Using a state-of-the-art simulation model, we can reproduce the individual-level dynamics of alcohol use and estimate changes associated with plausible policy reforms. This information is key for designing and advocating effective policy changes and advancing toward a population approach to prevention science.

C1. Specific aims

1. Select and describe specific alcohol policies based on their effectiveness, expected results, and feasibility of implementation in Chile in the next decade.
2. Generate a simulation model that mimics the empirical distribution, trend, and dynamics of alcohol use and related harm in the population of 15+ years of age of Chile between 2008 and 2019.
3. Estimate the 10-year impact of selected alcohol policies on alcohol use, cause-specific alcohol-related mortality, and years of potential life lost, across different demographic and socioeconomic groups, through a simulation model of the Chilean population.

D. METHODOLOGY

D1. Design:

We will build a microsimulation model to estimate the effect of different alcohol policies on alcohol use and harm. Microsimulation is a powerful modeling technique that can offer valuable information on public policies using multiple sources of data, including individual-level data. Policy effects are generated through the manipulation of conditions in which individuals from a (simulated) synthetic population behave under a variety of assumptions and scenarios.^{36,37} In our project, we will use the year 2008 as the starting year because of the consistency and comparability of alcohol data and restrict the analysis to the population of 15+ years of age, to be consistent with the different data sources we will use, and because this is the group in which the vast majority of alcohol-related harms are concentrated.²⁰ Model outputs will be reported for the entire population, as well across sex, age, and categories of educational attainment.

D2. Defining alcohol policy interventions (Specific aim 1)

Because it is impractical to simulate the effect of all alcohol policies in this project, we will use the list of WHO "best buys" to initiate a guided discussion with experts and key informants about policies that in

Table.1. Alcohol-related causes, source of relative risk, and ICD-10 codes

Causes	Source	ICD-10 Codes
Neuro-psychiatric		
Mental and behavioral disorders due to alcohol	PAF=1	F10
Degeneration of nervous system due to alcohol	PAF=1	G312
Epilepsy	Samokhvalov (2010) ³⁸	G40-G41
Alcohol polyneuropathy	PAF=1	G621
Cardiovascular		
Hypertensive disease	WHO (2018) ²⁰	I10-I15
Ischemic heart disease	Rehm (2016) ³⁹	I20-I25
Alcoholic cardiomyopathy	PAF=1	I426
Cardiac arrhythmias		I48X
Hemorrhagic and other non-ischemic stroke	Larsson (2016) ⁴⁰	I60-I62
Ischemic stroke	Rehm (2016) ³⁹	I63-I66
Cancer		
Mouth, oropharynx and laryngeal cancer	Bagnardi (2016) ⁴¹	C00-C14
Esophageal cancer	Bagnardi (2016) ⁴¹	C15
Colon cancer	Bagnardi (2016) ⁴¹	C18
Colorectal cancer	Bagnardi (2016) ⁴¹	C19X
Rectal cancer	Bagnardi (2016) ⁴¹	C20
Liver cancer	Turati (2014) ⁴²	C22
Laryngeal cancer	Bagnardi (2016) ⁴¹	C32
Breast cancer	Bagnardi (2016) ⁴¹	C50
Other causes attributable to alcohol		
Non-insulin-dependent diabetes mellitus	Knott (2015) ⁴³	E11
Tuberculosis	Imtiaz (2017) ⁴⁴	A15-A19
HIV/AIDS	Rehm (2017) ⁴⁵	B20-B24
Pneumonia	Samokhvalov (2010) ⁴⁶	J12-J18
Alcoholic gastritis	PAF=1	K292
Cirrhosis of the liver	WHO (2018) ²⁰	K70, K74
Alcohol-induced chronic pancreatitis	PAF=1	K860
Acute and chronic pancreatitis	Samokhvalov (2015) ⁴⁷	K85, K861
Injuries		
Intentional self-poisoning by and exposure to alcohol	PAF=1	X65
Accidental poisoning by and exposure to alcohol	PAF=1	X45
Poisoning by and exposure to alcohol, undetermined intent	PAF=1	Y15
Motor vehicle accidents	WHO (2018) ²⁰	a
Accidental poisoning	WHO (2018) ²⁰	X40, X46-X49
Falls	WHO (2018) ²⁰	W00-W19
Fires	WHO (2018) ²⁰	X00-X09
Drowning	WHO (2018) ²⁰	W65-W74
Other Unintentional injuries	WHO (2018) ²⁰	b
Self-inflicted injuries	WHO (2018) ²⁰	X60-X64, X66-X84, Y87.0
Homicide	WHO (2018) ²⁰	X85-Y09, Y87.1
Other intentional injuries	WHO (2018) ²⁰	Y35

PAF=Population attributable fraction. WHO=World Health Organization
a V021-V029, V031-V039, V041-V049, V092, V093, V123-V129, V133-V139, V143-V149, V194-V196, V203-V209, V213-V219, V223-V229, V233-V239, V243-V249, V253-V259, V263-V269, V273-V279, V283-V289, V294-V299, V304-V309, V314-V319, V324-V329, V334-V339, V344-V349, V354-V359, V364-V369, V374-V379, V384-V389, V394-V399, V404-V409, V414-V419, V424-V429, V434-V439, V444-V449, V454-V459, V464-V469, V474-V479, V484-V489, V494-V499, V504-V509, V514-V519, V524-V529, V534-V539, V544-V549, V554-V559, V564-V569, V574-V579, V584-V589, V594-V599, V604-V609, V614-V619, V624-V629, V634-V639, V644-V649, V654-V659, V664-V669, V674-V679, V684-V689, V694-V699, V704-V709, V714-V719, V724-V729, V734-V739, V744-V749, V754-V759, V764-V769, V774-V779, V784-V789, V794-V799, V803-V805, V811, V821, V830-V833, V840-V843, V850-V853, V860-V863, V870-V878, V892. b W20-W64, W 75-W99, X10-X39, X50-X59, Y40-Y86, Y88, Y89 and V series except for the codes listed in note a.

their opinion would be 1) more cost-effective in a social and epidemiological context like Chile, and 2) on their feasibility of being implemented in a 10-year period in the Chilean context based on social, political and fiscal constraints. For this, we will use a Delphi method, which seeks the opinion of a group of informants to assess the extent of agreement and to resolve disagreement on an issue.⁴⁸ We will start from a detailed list and explanation of alcohol policies included in the WHO "best buy" list, including a summary of the current policy status in Chile. In the first round we will ask informants to rank policies according to the two criteria mentioned above, giving them the opportunity to add one or more policies not included in the initial list. Main feasibility concerns will be elicited at this stage as an open-ended question. We will use an online questionnaire that automatically registers responses and generates a dataset (e.g., Google Forms). After analyzing and consolidating the information from the first round, we will initiate a second round of consultation to ask for their level of agreement with the first three policy preferences resulting from the initial consultation. Emergent information of the first round on feasibility concern will be presented to the respondents at this stage. In addition, in the second round, we will ask for their inputs on the specific regulation that the two selected policies should have in Chile. For example, if an increase/revision of alcohol taxation is selected, informants should expand on how they visualize this policy reform, including changes in the taxation scheme (e.g., *ad valorem* excise, specific excise, or other formula) and amount (e.g., increase in X% for spirits). It has been established that a Delphi panel needs a minimum of 30 experts to provide rigor for statistical analysis.⁴⁹ We propose a sample of 40 informants from academia, policy-makers, NGOs, and industry, expecting a minimum of 80% response rate in the first round.

D3. Data and data sources

To inform the distribution of sociodemographic characteristics, we will use the **Survey for National Socioeconomic Characterization of Chile (CASEN)**. CASEN is a nationally and regionally representative

survey that uses a probabilistic two stage random sample, where the primary sample unit are blocks or sections, depending on the year, and the second unit are households. We will use surveys from 2009, 2011, 2013, 2015, and 2017 to estimate the population distribution of sex, age, marital status, educational attainment, and employment status. CASEN microdata is publicly available and can be downloaded from the website of the Ministry of Social Development and Family of Chile (<https://shorturl.at/EH257>).

To estimate **transition probabilities across different demographic and socioeconomic groups**, we will use the **Social Protection Survey** (EPS for its acronym in Spanish), which is the largest and oldest longitudinal survey in Chile. It has a sample size of approximately 16,000 people distributed across all regions of the country. This survey contains detailed information on marital status, family history, educational attainment, and employment status, among several other sociodemographic dimensions. Our study will use the waves from 2009, 2012, 2015, and 2019 (all the available waves of the last decade). The EPS microdata are publicly available and can be requested from the website of the Vice Ministry of Social Protection of Chile (<https://shorturl.at/efpL0>).

To inform alcohol use behavior in the micro-synthetic population, we required an individual-level dataset containing sociodemographic variables along with information on **individuals' patterns of alcohol use**. For this, we used the **National Household Surveys on Drug Use of Chile** for 2008, 2010, 2012, 2014, 2016, and 2018. The survey is a nationally and regionally representative study and includes population between 15-64 years of age living in households in urban areas. It includes urban areas of 30,000+ inhabitants and uses a three-stage random sample stratified by municipality, where the primary sample unit are blocks or sections, depending on the year, the second unit are households and the third and final sampling unit is a household member within the eligible age group selected via a Kish table. The survey follows a standardized protocol for data collection, which includes a paper-and-pencil or computer assisted face-to-face interview, depending on the year of the survey, conducted by trained pollsters. The median sample size for the selected years was 18,151, and the response rates ranged between 62-89%. The National Household Surveys on Drug Use from Chile provide individual-level nationally representative repeated cross-sections. The survey contains information on respondents' biological sex, age, marital status, educational attainment, and employment status, among other sociodemographic variables. Alcohol use will be characterized based on the first three items of the *Alcohol Use Disorder Identification Tests (AUDIT)*, which measures drinking frequency, drinking quantity in a normal day, and frequency of binge drinking. We selected these items because they allow the estimation of the volume of alcohol use, because it has been validated in the Chilean population,⁵⁰ and because it has been included uninterruptedly in the survey's questionnaire since 2008. Unrestricted access to survey microdata is granted to the principal investigator of this proposal, according to the signed memorandum of understanding No. 27 from the year 2023 between the National Service for the Prevention and Rehabilitation of Drugs and Alcohol and Dr. Castillo-Carniglia (see the document in Annex section).

We will use **annual mortality datasets** that contain information for all deceased people between 2008 and 2019. We selected the period before the Covid-19 pandemic to avoid distortions in cause-specific mortality during the highest point of the pandemic. Mortality statistics include basic demographic information of the deceased, including sex, age, marital status, educational attainment, employment status, as well as the date, region, and municipality in which the death occurred, and the underlying and external cause of death coded according to ICD-10. Mortality statistics have improved substantially in the past two decades in Chile, (e.g., an increase in medical certifications for deaths occurred in rural areas, which are strongly correlated with ill-defined causes), and are considered of good quality in the global context.⁵¹ Access to mortality data is granted under the memorandum of understanding signed in April 2023 between the Vice Secretary of Public Health at the Chilean Ministry of and Universidad Mayor, in which the principal investigator of this proposal, Dr. Alvaro Castillo Carniglia is the designated counterpart (see the document in the Annex section).

D4. Variables

D4.1. Outcomes.

Alcohol use will be coded and reported in five categories, following the procedures and definition of prior research.^{2,4} Categories will be based on average grams of pure alcohol per day (g/day): abstainers (in the past 12 months); category 1, defined as $\leq 20/\leq 40$ g/day for women/men; category 2, defined as 21–40/41–60 g/day for women/men; category 3, defined as 41–60/61–100 g/day for women/men; and category 4, defined as $> 60/> 100$ g/day for women/men. As mentioned above, we will use the first three items of AUDIT. Grams per day will be calculated by applying the formula $V1_i = (F1_i - F2_i) * Q_i$ where $V1$ is the number of days consumed in a typical month, $F1$ is the number of days in the last month with alcohol consumption, $F2$ is the number of days with binge drinking episodes and Q is the number of standard drinks (i.e., amount of an alcoholic beverage with a total content of pure alcohol of 13.5 grams, e.g., a bottle of beer of 341cc with 5% of alcohol) consumed in a typical day. Subscript i denotes each observation in the survey. We used

the formula $V2_i = F2_i * 5$ to consider the number of drinks from binge episodes. The sum of $V1$ and $V2$, divided by 30 is equal to the average number of drinks consumed per day. As demonstrated widely in the alcohol literature, reports of alcohol use can be seriously underestimated; to deal with such problems, it is standard practice to triangulate^{52,53} the surveys' results with estimates of the volume of alcohol per capita estimated from statistics of production, imports, and exports in each country, plus an estimate of unrecorded alcohol use for the population aged 15 + years. To upshift the estimated volume of alcohol use, we will multiply the volume estimated from the survey by the inverse of the survey's coverage rate (i.e., survey estimates/per capita alcohol consumption). Then, the triangulated results will be recoded across categories of sex, age, and educational attainment according to the definition presented above, which will then be used as the population exposure to alcohol in the calculation of alcohol-attributable mortality and YLL.

Cause-specific alcohol-related mortality will be estimated by applying the specific alcohol population attributable fraction (PAF), using a modification of Levin's formula for multiple categories of exposure:⁵⁴ $PAF = \sum_i^k p_i(RR_i - 1) / \sum_i^k p_i(RR_i - 1) + 1$, where p is the prevalence for each category of alcohol consumption ($i=1 \dots 5$), sex and age group, and RR_i is the mortality relative risk for each cause associated with each level of exposure (relative to the abstainers category). Alcohol-related causes, their corresponding ICD-10 codes, and the source of the risk function for the PAF calculation are presented in **Table 1**. These estimates will be the basis of the estimated transition probabilities in the microsimulation model from each category of alcohol consumption to each specific cause of death.

Years of potential life lost (YPLL) will be estimated using the formula $\sum_i^k (reference_age - age_at_death_i)$ with i to k being the number of death attributable to alcohol consumption in people equal or younger to the reference life expectancies at birth in Chile in 2015-2020, which is 77.3 years for men and 82.1 for women.⁵⁵

D4.2. Exposure: alcohol policies scenarios.

As mentioned, our starting point will be the WHO "best buys",²⁰ namely: a) increase excise taxes on alcoholic beverages; b) enact and enforce bans or comprehensive restrictions on exposure to alcohol advertising (across multiple types of media); c) enact and enforce restrictions on the physical availability of retailed alcohol, specifically through reduced hours of sale; d) enact and enforce drink-driving laws and blood alcohol concentration limits via sobriety checkpoints; e) provide brief psychosocial intervention for persons with hazardous and harmful alcohol use; f) carry out regular reviews of prices in relation to the level of inflation and income; establish minimum prices for alcohol where applicable; enact and enforce an appropriate minimum age for purchase or consumption of alcoholic beverages and reduce density of retail outlets; g) restrict or ban promotions of alcoholic beverages in connection with sponsorships and activities targeting young people; h) provide prevention, treatment, and care for alcohol use disorders and comorbid conditions in health and social services; and i) provide consumer information about, and label, alcoholic beverages to indicate the harmful effects of alcohol.

Each of these policies has specific features that will be discussed with experts and informants, as explained for Aim 1. For example, reducing hours of sale depends on Chile's municipal ordinance and can be different for on- and off-premise outlets and for different types of alcohol patents; it includes regulation on closing hours, but also for opening and delivery. Its effectiveness is largely dependent on enforcement and associated sanctions. It is worth mentioning that the Chilean 2010 National Alcohol Strategy proposes concrete actions in some of the same areas; thus, it will be a valuable precedent in our discussion with experts on alcohol policies.

D4.3. Sociodemographic variables.

We will consider the following demographic and socioeconomic variables: biological sex (male or female), age (15-17, 18-24, 25-29, 30-34, 35-39, 40-44, 45-49, 50-59, 60-79, and 80+ years), marital status (single, married/cohabiting, divorced/separated, widower), educational attainment (less than primary education, primary or incomplete secondary education, secondary or incomplete technical/university education, graduate degree, postgraduate degree), and employment status (inactive, employed, unemployed). These variables were selected because they are the only common variables consistently reported across datasets.

D5. Analysis

The model we developed will assess the medium- and long-term effects of different relevant alcohol policies that the Chilean Government could implement and that emerge from the literature as potentially worthwhile options. The specific design for each modelled intervention will be co-designed with key informants (see Sections **D2. And D4.2**) and using as starting point the advances achieved in previous published alcohol policy models (the Sheffield Alcohol Policy Model, CASCADE, and SIMAH) but incorporating our population information, behavioral dynamics, and policy context. Our model will adopt a dynamic

microsimulation approach in which individuals are simulated over their lifespan, considering characteristics such as sex, age, and educational attainment.⁵⁶ These simulated individuals will conform a synthetic population. The model is generated through a system of differential equations that use empirical data to estimate transitions for each hypothetical individual across discrete behavioral (e.g., alcohol use) and health (e.g., mortality) states.

The models will be initialized with the data from the National Household Surveys on Drug Use of Chile. The dynamic component of the microsimulation derives from inputs generated from the data (or scientific literature if Chilean data is not available), specifically from our estimate of transition probabilities from one discrete state to another. For the socioeconomic variables measured longitudinally in the Social Protection Survey (educational attainment, employment status, and marital status), we will estimate transition probabilities using a multi-state model, assuming that states are concatenated following a Markovian or semi-Markovian process.⁵⁷ In multi-state models, transition probabilities can be defined as $P(Y(t) = b | Y(s) = a, H_{s-})$, where $a, b \in Z$, and where Z takes the values in the finite state space $[1, \dots, Z]$. $P(Y_t)$ is the probability of being in state b at time t , given that it was in state a at time s and conditional on the past trajectory until time s .⁵⁷ We propose using a parametric multi-state model that allows the specification of different distributions (e.g., exponential, Weibull, Gompertz) for each transition, making this approach a very flexible framework to model these data. The multistate model will be fitted using the flexsurv package in R.⁵⁸

Since we do not have longitudinal representative data on alcohol consumption, transition probabilities over time across alcohol categories for each sociodemographic group will be estimated using the method proposed by van de Dutta et al. for repeated cross-sectional studies (in our case, the National Household Surveys on Drug Use). With this method, transition probabilities are estimated using Langevin dynamics, which uses a set of stochastic differential equations that capture the temporal dynamics of specific data systems.⁵⁹ This method has shown significant predictive power when compared to longitudinal datasets. For this we will use the Langevine package in R.⁵⁹

The excess mortality due to alcohol consumption for female and male populations by specific category of sex, age, and educational attainment will be estimated by incorporating the cause-specific attributable fraction of mortality from each category of alcohol use, as explained in **Section D4.1**.

As in most microsimulation models, the effect of the different policy scenarios will be evidence oriented, meaning that we will use pooled estimates from meta-analysis or coefficient of regression models extracted from the literature (e.g., price elasticity of alcoholic beverages). The search for the best available information for this purpose requires a tradeoff analysis that incorporates the size and context of the pooled results, as well as their level of stratification across groups of interests (e.g., sex, age group, and category of educational attainment). The more detailed the effect estimates, the more information we can incorporate into the model.

There are multiple programming languages and software to generate a microsimulation model (e.g., R, Python, JAVA, and C++). Here, we propose to use the MicSim package in R, which is a discrete-state simulation model, in which the simulated population is open, meaning that individuals can enter (e.g., individuals below the age of 15 that were excluded from the population can turn 15 during model runs) or leave (e.g., due to death) the population, and life-courses are specified to evolve along continuous time scales (e.g., calendar time).⁶⁰ Mathematically, an individual life-course can be characterized as a stochastic (usually Markovian) process $Z(t)$, $t \in R_0^+$.⁶⁰ Here, the process time t represents the time span over which simulated individual evolves. Time t is set to zero when an individual enters the virtual population. As long as an individual is part of the synthetic population, t evolves along the individual lifetime.⁶⁰

All simulation models have several layers of uncertainty (from random variability due to chance, uncertainty in parameter estimation and model assumptions). **We will follow international recommendations to estimate and report the uncertainty on our outcomes.**⁶¹ Parameter uncertainty will be accounted assuming a continuous distribution for each estimated parameter based on the point estimate (mean) and the precision (standard error) including it in a probabilistic sensitivity analysis.^{61,62} Additionally, we will check the robustness of our results against changes in parameter estimates for model building, including information to estimate alcohol consumption (e.g., % of unrecorded alcohol use, % of alcohol consumed from per capita estimates) and mortality risk parameters for former consumers.⁵²

D6. Ethical considerations.

We will submit our project to the Universidad Mayor's Scientific Ethics Committee (accredited by Araucania's Health Regional Secretary under Administrative Act No. J1-6888 of December 15th, 2020). All files will be stored in a HIPAA (U.S. Health Insurance Portability and Accountability Act) complying cloud storage, such as Google Workspace, Atlantic.Net, Amazon S3, or similar, which encrypts and keeps all files in secure storage servers. In addition, all results will be presented in aggregate form (e.g., effect across population subgroups) so that individual profiles and trajectories will not be revealed at any moment. Any further recommendation raised by the Scientific Ethics Committee will be incorporated.

E. NOVELTY

Our proposal advances several areas of scientific research. **First**, to the best of our knowledge, this is the first study in Latin America to use microsimulation to estimate the effects of alcohol policies. There have been prior efforts to adapt existing microsimulation models to examine hypothetical changes in social policies, including basic solidarity old age pension or abolishing deductions from personal expenditures from personal income tax,⁶³ but none of these are specific for alcohol and do not consider health outcomes. **Second**, we will use a state-of-the-art methodology for simulating health, social, and economic policies, taking advantage of the current developments in computational modelling. Microsimulation offers several advantages for policy analysis, including (i) a detailed understanding of how policy changes affect individuals and subgroups within the population, allowing policymakers to design targeted interventions; (ii) it will help identify unintended consequences or policy spillovers that may arise from complex interactions between individuals; and (iii) it allows for scenario testing and sensitivity analysis, enabling policymakers to explore different policy options and assess their robustness. **Third**, we will combine and use the best available data from Chile and abroad, including vital and administrative data, survey microdata, results from policy evaluation, and meta-analysis on the effect of alcohol use on health. As part of the microsimulation model, we will also produce updated statistics on alcohol-attributable mortality and years of potential life lost in Chile as well as a comprehensive analysis of alcohol use to generate and describe the alcohol dynamics of the population, including estimating transition probabilities from discrete states to another (e.g., from the abstainer to moderate alcohol use categories). **Fourth**, our project uses an interdisciplinary (computer science, economics, public policy, public health, and epidemiology) and system approach to study a complex problem that has differential impact across population subgroups, particularly among more disadvantage groups.⁴

F. RELEVANCE AND IMPACT

Several aspects of our proposal ensure its relevance and impact. **First**, as mentioned in the background section, it covers one of the leading risk factors for morbidity and mortality in Chile and most other countries in the world. As such, its results, although specific to the Chilean population, can be used in contexts outside the Global North, where most of the research on this topic is produced. **Second**, it directly informs policy discussions that could positively affect multiple aspects of social living, including life expectancy, quality of life, health expenditure, social development, and national sanitary goals for the decade.⁶⁴ The latter is sustained, in part, on the flexibility provided by the microsimulation model in which output can be easy to understand and communicate for large audiences. For example, outputs can be expressed as absolute changes in alcohol prevalence or the number/rate/proportion of deaths that could be avoided with the implementation of one policy over the 10 years following implementation. The latter can make the discussion process more efficient by focusing on policies with the largest potential health impact in a context such as Chile. **Third**, the model setup can be extended to estimate the effect of additional alcohol policies on other outcomes (economic costs, other social externalities), and include other behavioral risk factors and health problems, such as tobacco use, diet, or physical activity. **Fourth**, in our commitment to transparency and reproducibility, we will pre-register the protocol of our project in an open-source repository such as the Open Science Foundation (<https://osf.io>) and make available all codes generated for data preparation, analysis, and simulation in a GitHub repository of the project. This is in line with international efforts to make science reproducible and to allow the adaptation of our model to respond to other research questions.

H. WORK PLAN AND RESEARCH TEAM

Based in our own experience with similar projects and the information provided by our international collaborators, the data acquisition, harmonization, and analysis for model setup and calibration is expected to be very demanding and time consuming. For example, the CASCADE project was a 5-year R01 grant and the SIMAH was a 3-year project with an administrative supplement for one additional year; in these two projects, multiple coders and data analysts were involved. Accordingly, we propose a 4-year project with activities detailed in **Table 2**. Several products will be generated in our project. We expect to publish **at least three scientific articles in top journals** in the substance abuse or public health fields. We will also produce reports and policy briefs for each policy analyzed, and present intermediate and final results in scientific meetings and seminars for the general population and for experts and stakeholders in this area, taking advantage of our current agreements with SENDA and the Ministry of Health. These seminars will include presentations and discussions with our international collaborators (see description below).

Our team has a strong publication record and years of experience conducting research on alcohol use and related harm, generating primary data, and using secondary information to evaluate and simulate policy effects. **Dr. Alvaro Castillo-Carniglia** is a social epidemiologist with vast experience in studying the epidemiology of substance use in Chile and abroad (e.g., Latin America and the US). Dr. Castillo-Carniglia

specializes in survey methodology,⁶⁵ policy evaluation,⁶⁶⁻⁶⁸ and use of quasi-experimental methods in general.^{69,70} Aligned with the goal of this proposal, his projects and publications include the analysis of the causes, consequences, and policy impacts in substance use and harms.^{66,71-73} He has formal training in complex system modelling and published a paper that used an agent-based model to simulate the effect of changing the density and distribution of alcohol outlets on alcohol use and related-violent events.⁷⁴ He has been the principal investigator/director of multiple studies to characterize and measure trends of substance use and related problems, including the National Drug Surveys in School and General Population, project FONDECYT regular 1191282 about trajectories of people in drug treatments in Chile modeled using multi-state models, and the Millennium Nucleolus for the Evaluation and Analysis of Drug Policy (NCS2021_003).

Dr. Cristóbal Cuadrado is a public health specialist and health economist with extensive research experience in Chile and the UK. Dr. Cuadrado has experience conducting impact evaluations of policy interventions using different quantitative methods applied to policy evaluation, including health system interventions,⁷⁵ interventions at the population level such as the tax on sugary drinks in Chile,^{76,77} population effects of the pandemic on access to services,^{78,79} cluster-randomized trials for community interventions in schools.⁸⁰ He also has experience in the development of decision models for economic evaluations to assess the cost-effectiveness of population public policies using Markov models and microsimulation leading projects to evaluate nutritional policy interventions, such as fiscal policies (Newton-Picarte MR/N026640/1), cost of obesity consequences in Chile (FONIS SA14ID0176) and the cost-effectiveness of different gastric cancer prevention strategies in Chile (FONIS SA19I0188).

In addition, **Dr. Robin Purshouse**, the principal investigator of the CASCADE model, and one of the main architects of the Sheffield Alcohol Policy Model and the Simulation of Alcohol Control Policies for Health Equity (SIMAH), and **Dr. Magdalena Cerdá**, principal investigator of the NIAAA R21 project “Neighborhood interventions in alcohol-related homicide: a systems approach,” which used an agent-based model of New York City to model different alcohol policies, have agreed to be international collaborators in our proposal. These collaborations ensure that the theoretical, technical, and practical aspects of our simulation model will be correctly executed (see the **letter of Dr. Purshouse and Cerdá in the Annex** of this proposal).

Table 2. Project’s timeline																
	Year 1				Year 2				Year 3				Year 4			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Aim 1																
Document with summary of WHO alcohol “best buys”																
Application, analysis, and writing of Delphi results																
Aim 2																
Collect and harmonize data sources																
Estimate sociodemographic parameters and dynamics																
Estimate alcohol parameters and dynamics																
Estimate mortality and YPLL																
Model development and validation																
Conference presentation 1																
Paper 1. The Chilean alcohol microsimulation model																
Aim 3 and 4																
Forecast alcohol use dynamics and related harms																
Specify and model the effect of policy changes																
Conference presentation 2 and 3																
FONDECYT progress report																
Paper 2 and 3. Policy effects on alcohol use/harm																
Policy briefs and seminar to present study results																
FONDECYT final report																

J. FUTURE DIRECTIONS

After achieving the project goals, we plan to augment the model by adding a layer of information on costs (from a societal perspective) to estimate the economic consequences of alcohol consumption, including direct and indirect costs. In addition, owing to the flexibility of the microsimulation model, we aimed to test different influential factors and their contribution to explaining consumption dynamics and related harms. This, for example, will allow us to test causal mechanisms through which alcohol policies may be operating and to know how policy design may potentiate their expected effects. Finally, taking advantage of our work with these methods,^{66,81} we plan to include a spatial component in the modelling process by generating smoothed estimates at the regional and municipal levels. Increasing the spatial resolution of the model estimates is of great value for local governments, ultimately responsible for the implementation or enforcement of policies and prevention strategies.

BIBLIOGRAPHIC REFERENCES:

1. Margozzini P, Paraje G, Zitko P, et al. *Estudio del costo economico y social del consumo de alcohol en Chile*. 2018. https://medicina.uc.cl/wp-content/uploads/2019/03/COSTO-ALCOHOL_Actualizacio%CC%81n-2018_Informe.pdf
2. Castillo-Carniglia A, Kaufman JS, Pino P. Alcohol-attributable mortality and years of potential life lost in Chile in 2009. *Alcohol Alcohol*. Nov-Dec 2013;48(6):729-36. doi:10.1093/alcalc/agt066
3. Brennan A, Buckley C, Vu TM, et al. Introducing CASCADEPOP: an open-source sociodemographic simulation platform for us health policy appraisal. *Int J Microsimul*. Summer 2020;13(2):21-60. doi:10.34196/ijm.00217
4. Probst C, Buckley C, Lasserre AM, et al. Simulation of Alcohol Control Policies for Health Equity (SIMAH) Project: Study Design and First Results. *Am J Epidemiol*. May 5 2023;192(5):690-702. doi:10.1093/aje/kwad018
5. Brennan A, Meier P, Purshouse R, Rafia R, Meng Y, Hill-Macmanus D. Developing policy analytics for public health strategy and decisions-the Sheffield alcohol policy model framework. *Ann Oper Res*. 2016;236(1):149-176. doi:10.1007/s10479-013-1451-z
6. Collaborators GBDRF. Global burden of 87 risk factors in 204 countries and territories, 1990-2019: a systematic analysis for the Global Burden of Disease Study 2019. *Lancet*. Oct 17 2020;396(10258):1223-1249. doi:10.1016/S0140-6736(20)30752-2
7. Rehm J, Shield KD. Global Burden of Alcohol Use Disorders and Alcohol Liver Disease. *Biomedicines*. Dec 13 2019;7(4)doi:10.3390/biomedicines7040099
8. Carvalho AF, Heilig M, Perez A, Probst C, Rehm J. Alcohol use disorders. *Lancet*. Aug 31 2019;394(10200):781-792. doi:10.1016/S0140-6736(19)31775-1
9. Castillo-Carniglia A, Keyes KM, Hasin DS, Cerda M. Psychiatric comorbidities in alcohol use disorder. *Lancet Psychiatry*. Dec 2019;6(12):1068-1080. doi:10.1016/S2215-0366(19)30222-6
10. Whiteford HA, Degenhardt L, Rehm J, et al. Global burden of disease attributable to mental and substance use disorders: findings from the Global Burden of Disease Study 2010. *Lancet*. Nov 9 2013;382(9904):1575-86. doi:10.1016/S0140-6736(13)61611-6
11. Karriker-Jaffe KJ, Room R, Giesbrecht N, Greenfield TK. Alcohol's Harm to Others: Opportunities and Challenges in a Public Health Framework. *J Stud Alcohol Drugs*. Mar 2018;79(2):239-243. doi:10.15288/jsad.2018.79.239
12. Room R, Callinan S, Greenfield TK, et al. The social location of harm from others' drinking in 10 societies. *Addiction*. Mar 2019;114(3):425-433. doi:10.1111/add.14447
13. Cherpitel CJ, Ye Y, Monteiro M. Risk of violence-related injury from alcohol consumption and its burden to society in Latin America and the Caribbean. *Rev Panam Salud Publica*. Jan 2018;42doi:10.26633/RPSP.2018.7
14. Rehm J, Manthey J, Lange S, et al. Alcohol control policy and changes in alcohol-related traffic harm. *Addiction*. Apr 2020;115(4):655-665. doi:10.1111/add.14796
15. Manthey J, Hassan SA, Carr S, Kilian C, Kuitunen-Paul S, Rehm J. What are the Economic Costs to Society Attributable to Alcohol Use? A Systematic Review and Modelling Study. *Pharmacoeconomics*. Jul 2021;39(7):809-822. doi:10.1007/s40273-021-01031-8
16. Gavurova B, Tarhanicova M. Methods for Estimating Avoidable Costs of Excessive Alcohol Consumption. *Int J Environ Res Public Health*. May 7 2021;18(9)doi:10.3390/ijerph18094964
17. World Health Organization. *Global and regional estimates of violence against women: prevalence and health effects of intimate partner violence and non-partner sexual violence*. 2013. <https://apps.who.int/iris/handle/10665/85239>
18. Boyd J, Sexton O, Angus C, Meier P, Purshouse RC, Holmes J. Causal mechanisms proposed for the alcohol harm paradox-a systematic review. *Addiction*. Jan 2022;117(1):33-56. doi:10.1111/add.15567
19. Bloomfield K. Understanding the alcohol-harm paradox: what next? *Lancet Public Health*. Jun 2020;5(6):e300-e301. doi:10.1016/S2468-2667(20)30119-5
20. World Health Organization. *Global status report on alcohol and health 2018*. 2018. <https://apps.who.int/iris/handle/10665/274603>
21. Servicio Nacional para la Prevención y Rehabilitación del Consumo de Drogas y Alcohol. *Décimo Cuarto Estudio Nacional de Drogas en Población General*. 2020. <https://www.senda.gob.cl/wp-content/uploads/2022/01/Estudio-PG2020.pdf>
22. Servicio Nacional para la Prevención y Rehabilitación del Consumo de Drogas y Alcohol. *Estrategia Nacional de Drogas 2021-2030. Para un Chile que Elige Vivir sin Drogas*. 2021. <https://www.senda.gob.cl/wp-content/uploads/2021/02/Estrategia-Nacional-de-Drogas-version-web.pdf>
23. Pena S, Sierralta P, Norambuena P, Leyton F, Pemjean A, Roman F. Alcohol policy in Chile: a

- systematic review of policy developments and evaluations. *Addiction*. Mar 2021;116(3):438-456. doi:10.1111/add.15208
24. Ministerio de Salud. *Estudio de carga de enfermedad y carga atribuible*. 2008. https://medicina.uc.cl/wp-content/uploads/2018/08/Informe_final_estudio_carga_2007.pdf
25. Florenzano R, Guzman E, Sieverson C, et al. [The harm to others from drinking: A population study]. *Rev Med Chil*. Oct 2015;143(10):1242-51. Dano a terceros producido por el alcohol: resultados de un estudio poblacional en Chile. doi:10.4067/S0034-98872015001000002
26. WHO/Thai Health International Collaborative Research Project. *The Harm to Others from Drinking: Master Protocol*. 2012.
27. Subsecretaría del Interior. Ley 19.925 sobre expendio y consumo de bebidas alcoholicas. Biblioteca del Congreso Nacional; 2004.
28. Ministerio del Interior y Seguridad Pública. Ley 21.363. Establece normas en materia de comercialización y publicidad de bebidas alcohólicas, y otras que indica. Biblioteca del Congreso Nacional; 2021.
29. Ministerio de Hacienda. Ley 20.780. Reforma tributaria que modifica el sistema de tributación de la renta e introduce diversos ajustes en el sistema tributario. Biblioteca del Congreso Nacional; 2014.
30. Roche M, Sandoval RC, Monteiro MG. Comparing taxes on alcoholic beverages in the Region of the Americas. *Addiction*. Jul 2023;118(7):1389-1395. doi:10.1111/add.16146
31. Holmes J, Meng Y, Meier PS, et al. Effects of minimum unit pricing for alcohol on different income and socioeconomic groups: a modelling study. *Lancet*. May 10 2014;383(9929):1655-1664. doi:10.1016/S0140-6736(13)62417-4
32. Purshouse RC, Brennan A, Rafia R, et al. Modelling the cost-effectiveness of alcohol screening and brief interventions in primary care in England. *Alcohol Alcohol*. Mar-Apr 2013;48(2):180-8. doi:10.1093/alcalc/ags103
33. Meier PS, Holmes J, Brennan A, Angus C. Alcohol policy and gender: a modelling study estimating gender-specific effects of alcohol pricing policies. *Addiction*. Sep 2021;116(9):2372-2384. doi:10.1111/add.15464
34. Buckley C, Field M, Vu TM, et al. An integrated dual process simulation model of alcohol use behaviours in individuals, with application to US population-level consumption, 1984-2012. *Addict Behav*. Jan 2022;124:107094. doi:10.1016/j.addbeh.2021.107094
35. McGill E, Petticrew M, Marks D, McGrath M, Rinaldi C, Egan M. Applying a complex systems perspective to alcohol consumption and the prevention of alcohol-related harms in the 21st century: a scoping review. *Addiction*. Sep 2021;116(9):2260-2288. doi:10.1111/add.15341
36. Krijkamp EM, Alarid-Escudero F, Enns EA, Jalal HJ, Hunink MGM, Pechlivanoglou P. Microsimulation Modeling for Health Decision Sciences Using R: A Tutorial. *Med Decis Making*. Apr 2018;38(3):400-422. doi:10.1177/0272989X18754513
37. Kim D. *New horizons in modeling and simulation for social epidemiology and public health*. Wiley series in modeling and simulation. Wiley; 2021.
38. Samokhvalov AV, Irving H, Mohapatra S, Rehm J. Alcohol consumption, unprovoked seizures, and epilepsy: a systematic review and meta-analysis. *Epilepsia*. Jul 2010;51(7):1177-84. doi:EPI2426 [pii] 10.1111/j.1528-1167.2009.02426.x
39. Rehm J, Shield KD, Roerecke M, Gmel G. Modelling the impact of alcohol consumption on cardiovascular disease mortality for comparative risk assessments: an overview. *BMC Public Health*. Apr 28 2016;16:363. doi:10.1186/s12889-016-3026-9
40. Larsson SC, Wallin A, Wolk A, Markus HS. Differing association of alcohol consumption with different stroke types: a systematic review and meta-analysis. *BMC Med*. Nov 24 2016;14(1):178. doi:10.1186/s12916-016-0721-4
41. Bagnardi V, Rota M, Botteri E, et al. Alcohol consumption and site-specific cancer risk: a comprehensive dose-response meta-analysis. *Br J Cancer*. Feb 3 2015;112(3):580-93. doi:10.1038/bjc.2014.579
42. Turati F, Galeone C, Rota M, et al. Alcohol and liver cancer: a systematic review and meta-analysis of prospective studies. *Ann Oncol*. Aug 2014;25(8):1526-35. doi:10.1093/annonc/mdl020
43. Knott C, Bell S, Britton A. Alcohol Consumption and the Risk of Type 2 Diabetes: A Systematic Review and Dose-Response Meta-analysis of More Than 1.9 Million Individuals From 38 Observational Studies. *Diabetes Care*. Sep 2015;38(9):1804-12. doi:10.2337/dc15-0710
44. Imtiaz S, Shield KD, Roerecke M, Samokhvalov AV, Lonnroth K, Rehm J. Alcohol consumption as a risk factor for tuberculosis: meta-analyses and burden of disease. *Eur Respir J*. Jul 2017;50(1)doi:10.1183/13993003.00216-2017

45. Rehm J, Probst C, Shield KD, Shuper PA. Does alcohol use have a causal effect on HIV incidence and disease progression? A review of the literature and a modeling strategy for quantifying the effect. *Popul Health Metr.* Feb 10 2017;15(1):4. doi:10.1186/s12963-017-0121-9
46. Samokhvalov AV, Irving HM, Rehm J. Alcohol consumption as a risk factor for pneumonia: a systematic review and meta-analysis. *Epidemiol Infect.* Dec 2010;138(12):1789-95. doi:S0950268810000774 [pii] 10.1017/S0950268810000774
47. Samokhvalov AV, Rehm J, Roerecke M. Alcohol Consumption as a Risk Factor for Acute and Chronic Pancreatitis: A Systematic Review and a Series of Meta-analyses. *EBioMedicine.* Dec 2015;2(12):1996-2002. doi:10.1016/j.ebiom.2015.11.023
48. Vogel C, Zwolinsky S, Griffiths C, Hobbs M, Henderson E, Wilkins E. A Delphi study to build consensus on the definition and use of big data in obesity research. *Int J Obes (Lond).* Dec 2019;43(12):2573-2586. doi:10.1038/s41366-018-0313-9
49. Chuenjitwongsa S. Conduct a Delphi Study. 2017. https://www.cardiff.ac.uk/_data/assets/pdf_file/0010/1164961/how_to_conduct_a_delphistudy.pdf
50. Alvarado ME, Garmendia ML, Acuna G, Santis R, Arteaga O. [Assessment of the alcohol use disorders identification test (AUDIT) to detect problem drinkers]. *Rev Med Chil.* Nov 2009;137(11):1463-8. Validez y confiabilidad de la version chilena del Alcohol Use Disorders Identification Test (AUDIT).
51. Mahapatra P, Shibuya K, Lopez AD, et al. Civil registration systems and vital statistics: successes and missed opportunities. *Lancet.* Nov 10 2007;370(9599):1653-63. doi:10.1016/S0140-6736(07)61308-7
52. Shield K, Mantney J, Rylett M, et al. National, regional, and global burdens of disease from 2000 to 2016 attributable to alcohol use: a comparative risk assessment study. *Lancet Public Health.* Jan 2020;5(1):e51-e61. doi:10.1016/S2468-2667(19)30231-2
53. Stockwell T, Zhao J, Sher K, Rehm J, Shield K, Naimi T. Underestimation of alcohol consumption in cohort studies and implications for alcohol's contribution to the global burden of disease. *Addiction.* Dec 2018;113(12):2245-2249. doi:10.1111/add.14392
54. Walter SD. The estimation and interpretation of attributable risk in health research. *Biometrics.* Dec 1976;32(4):829-49.
55. Instituto Nacional de Estadísticas. Esperanza de vida. Accessed 06/07, 2023. <https://www.inec.cl/ine-ciudadano/definiciones-estadisticas/poblacion/esperanza-de-vida#:~:text=En%20Chile%20la%20Esperanza%20de,77%2C3%20para%20los%20hombres>.
56. Breeze PR, Squires H, Ennis K, et al. Guidance on the use of complex systems models for economic evaluations of public health interventions. *Health Econ.* Jul 2023;32(7):1603-1625. doi:10.1002/hec.4681
57. Crowther MJ, Lambert PC. Parametric multistate survival models: Flexible modelling allowing transition-specific distributions with application to estimating clinically useful measures of effect differences. *Stat Med.* Dec 20 2017;36(29):4719-4742. doi:10.1002/sim.7448
58. Jackson CH. flexsurv: A Platform for Parametric Survival Modeling in R. *J Stat Softw.* May 12 2016;70doi:10.18637/jss.v070.i08
59. Rinn P, Lind P, Wächter M, Peinke J. The Langevin Approach: An R Package for Modeling Stochastic Processes. *Journal of Open Research Software.* 2016;4(1):e34. doi:10.5334/jors.123
60. Zinn S. The MicSim package of R: An entry-level toolkit for continuous-time microsimulation. *International Journal of Microsimulation.* 2014;7(3):3-32. doi:10.34196/IJM.00105
61. Briggs AH, Weinstein MC, Fenwick EA, et al. Model parameter estimation and uncertainty: a report of the ISPOR-SMDM Modeling Good Research Practices Task Force--6. *Value Health.* Sep-Oct 2012;15(6):835-42. doi:10.1016/j.jval.2012.04.014
62. Baio G, Dawid AP. Probabilistic sensitivity analysis in health economics. *Stat Methods Med Res.* Dec 2015;24(6):615-34. doi:10.1177/0962280211419832
63. Collado D, Contreras D, Duarte F, Gómez-Lobo A, Sanhueza C. CHILMOD. Centre for Microsimulation and Policy Analysis. Accessed 06/06, 2023. <https://www.microsimulation.ac.uk/euromod/models/chilmod/>
64. Ministerio de Salud. Estrategia nacional de salud para los objetivos sanitarios al 2030. 2022. <https://www.minsal.cl/wp-content/uploads/2022/03/Estrategia-Nacional-de-Salud-2022-MINSAL-V8.pdf>
65. Castillo-Carniglia A, Pizarro E, Marin JD, Rodriguez N, Casas-Cordero C, Cerda M. The effects of using answer sheets on reported drug use and data quality in a classroom survey: A cluster-randomized study. *Drug Alcohol Depend.* Sep 1 2017;178:194-200. doi:10.1016/j.drugalcdep.2017.05.012
66. Castillo-Carniglia A, Rivera-Aguirre A, Santaella-Tenorio J, et al. Changes in Opioid and Benzodiazepine Poisoning Deaths After Cannabis Legalization in the US: A County-level Analysis, 2002-2020. *Epidemiology.* Jul 1 2023;34(4):467-475. doi:10.1097/EDE.0000000000001609
67. Castillo-Carniglia A, Gonzalez-Santa Cruz A, Cerda M, et al. Changes in opioid prescribing after implementation of mandatory registration and proactive reports within California's prescription drug

- monitoring program. *Drug Alcohol Depend.* Jan 1 2021;218:108405. doi:10.1016/j.drugalcdep.2020.108405
68. Laqueur H, Rivera-Aguirre A, Shev A, et al. The impact of cannabis legalization in Uruguay on adolescent cannabis use. *Int J Drug Policy.* Jun 2020;80:102748. doi:10.1016/j.drugpo.2020.102748
69. Castillo-Carniglia A, Kagawa RMC, Cerda M, et al. California's comprehensive background check and misdemeanor violence prohibition policies and firearm mortality. *Ann Epidemiol.* Feb 2019;30:50-56. doi:10.1016/j.annepidem.2018.10.001
70. Castillo-Carniglia A, Ponicki WR, Gaidus A, et al. Prescription Drug Monitoring Programs and Opioid Overdoses: Exploring Sources of Heterogeneity. *Epidemiology.* Mar 2019;30(2):212-220. doi:10.1097/EDE.0000000000000950
71. Castillo-Carniglia A, Kaufman JS, Pino P. Geographical distribution of alcohol-attributable mortality in Chile: a Bayesian spatial analysis. *Addict Behav.* Mar 2015;42:207-15. doi:10.1016/j.addbeh.2014.11.025
72. Castillo-Carniglia A, Pizarro E, Luengo D, Soto-Brandt G. [Alcohol use and self-reported violent events in Chile]. *Adicciones.* 2014;26(1):46-53. Consumo de alcohol y autoinforme de eventos violentos en Chile.
73. Castillo-Carniglia A, Rivera-Aguirre A, Calvo E, Queirolo R, Keyes KM, Cerda M. Trends in marijuana use in two Latin American countries: an age, period and cohort study. *Addiction.* Nov 2020;115(11):2089-2097. doi:10.1111/add.15058
74. Castillo-Carniglia A, Pear VA, Tracy M, Keyes KM, Cerda M. Limiting Alcohol Outlet Density to Prevent Alcohol Use and Violence: Estimating Policy Interventions Through Agent-Based Modeling. *Am J Epidemiol.* Apr 1 2019;188(4):694-702. doi:10.1093/aje/kwy289
75. Pacheco J, Cuadrado C, Martinez-Gutierrez MS. Urgent care centres reduce emergency department and primary care same-day visits: a natural experiment. *Health Policy Plan.* Apr 1 2019;34(3):170-177. doi:10.1093/heapol/czz023
76. Nakamura R, Mirelman AJ, Cuadrado C, Silva-Illanes N, Dunstan J, Suhrcke M. Evaluating the 2014 sugar-sweetened beverage tax in Chile: An observational study in urban areas. *PLoS Med.* Jul 2018;15(7):e1002596. doi:10.1371/journal.pmed.1002596
77. Cuadrado C, Dunstan J, Silva-Illanes N, Mirelman AJ, Nakamura R, Suhrcke M. Effects of a sugar-sweetened beverage tax on prices and affordability of soft drinks in Chile: A time series analysis. *Soc Sci Med.* Jan 2020;245:112708. doi:10.1016/j.socscimed.2019.112708
78. Pacheco J, Crispi F, Alfaro T, Martinez MS, Cuadrado C. Gender disparities in access to care for time-sensitive conditions during COVID-19 pandemic in Chile. *BMC Public Health.* Oct 19 2021;21(1):1802. doi:10.1186/s12889-021-11838-x
79. Cuadrado C, Vidal F, Pacheco J, Flores-Alvarado S. Cancer Care Access in Chile's Vulnerable Populations During the COVID-19 Pandemic. *Am J Public Health.* Aug 2022;112(S6):S591-S601. doi:10.2105/AJPH.2021.306587
80. Pena S, Carranza M, Cuadrado C, et al. Effectiveness of a Gamification Strategy to Prevent Childhood Obesity in Schools: A Cluster Controlled Trial. *Obesity (Silver Spring).* Nov 2021;29(11):1825-1834. doi:10.1002/oby.23165
81. Castillo-Carniglia A, Kaufman JS, Pino P. Small area associations between social context and alcohol-attributable mortality in a middle income country. *Drug Alcohol Depend.* Apr 1 2014;137:129-36. doi:10.1016/j.drugalcdep.2014.01.020